

8	(a)	(i)	Polarisation is said to occur when particles in the wave <u>oscillate only in a single direction perpendicular to the direction of energy transfer</u> of the wave.	A1
		(ii)	1. $I_0 = 2 I_1$	A1
			2. $I_2 = I_1 \cos^2 20^\circ = 0.883 I_1$ $I_3 = I_2 \cos^2 30^\circ = 0.662 I_1 \approx 0.66 I_1$ (2 or 3 s.f.)	C1 A1
	(b)	(i)	Coherent: Observable interference pattern or maximas and minimas Same amplitudes: Minima is at zero intensity (Accept: complete cancellation)	B1 B1
		(ii)	$x = \frac{14.8 - 5.2}{6} = 1.6 \text{ mm}$ (Accept: $x = 1.6 \text{ mm}$ directly stated or substituted in equation below.) $a = \frac{\lambda D}{x} = \frac{650 \times 10^{-9} \times 1.5}{1.6 \times 10^{-3}}$ $a = 0.61 \text{ mm}$	C1 M1 A1
		(iii)	One wavelength (λ) or 650 nm	A1
		(iv)	$\Delta\phi = \frac{1.4}{1.6} \times 2\pi$ $= 5.5 \text{ rad}$	C1 A1
		(v)	$\theta = \tan^{-1}\left(\frac{0.0096}{1.5}\right) = 0.0064 \text{ rad}$ $b = \frac{\lambda}{\sin\theta} = \frac{650 \times 10^{-9}}{\sin(0.0064 \text{ rad})}$ $b = 0.102 \text{ mm}$	C1 M1 A1
	(c)	(i)	$d \sin \theta_n = n\lambda$ Since d and $\sin \theta_n$ are the same when the fringes overlap, $n\lambda = \text{constant}$. $\frac{\lambda_{\text{original}}}{\lambda_M} = \frac{3}{2} \Rightarrow \lambda_M = \frac{2(650 \times 10^{-9})}{3}$ $\lambda_M = 433 \times 10^{-9} \text{ m} = 433 \text{ nm}$ Therefore, light M is indigo. (Accept violet or blue)	C1 M1 A1
		(ii)	<u>Central</u> bright fringe is <u>white</u> . The rest of the bright fringes/ <u>others</u> contain a <u>spectrum</u> of colours.	B1 B1

9	(a)	1) The existence of a threshold frequency below which no photoelectrons are
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